Integrating Explainable AI with an LLM-Based Module for Household Energy Load Scheduling

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***Abstract*—Residential energy management systems require decision support tools that are both accurate and transparent, yet existing machine learning approaches for energy consumption forecasting are inherently opaque, and existing LLM-based scheduling systems lack grounding in data-driven forecasting evidence. This paper presents a framework integrating SHAP-based explainability from a residential energy consumption forecasting model with a single LLM-based scheduling module for 24-hour household appliance load scheduling. An XGBoost model trained on real residential energy data from the PECAN Street Austin dataset achieves a test R^2 of 0.651. SHAP TreeExplainer is applied to generate structured feature attribution summaries, which are passed to a Llama 3 scheduling module as decision context under two experimental conditions, with and without SHAP context, enabling a controlled comparison across a seven-day evaluation period within the held-out test set. A key empirical finding is that electricity price has near-zero predictive importance in passive household consumption, providing data-driven motivation for active scheduling intervention. The evaluation demonstrated that SHAP context produced more grounded explanations, with Condition A achieving a higher explanation completeness score and referencing solar generation more consistently than Condition B. Scheduling performance demonstrated clear improvement under Condition A, achieving a mean cost saving of 71.0%. Results are discussed in the context of prompt design limitations and directions for future work.**

***Index Terms*—explainable AI, SHAP, XGBoost, large language models, home energy management, load scheduling**

I. Introduction

The increasing penetration of renewable energy resources and the rise in time-of-use electricity tariffs have created opportunities for demand-side flexibility in residential energy management. Households equipped with rooftop solar systems can reduce electricity costs by shifting appliance usage to periods of high solar generation or low grid prices [1]. However, realising this potential requires decision support tools that are both accurate and transparent, capable not only of recommending optimal schedules but also of explaining why these recommendations are made in terms meaningful to the user.

Machine learning models and gradient boosted tree ensembles have demonstrated strong performance in residential energy forecasting [2], [3]. However, such models are inherently opaque; accurate predictions are produced without any insight into the features driving them. This black-box nature represents a challenge in energy management applications. As model complexity increases to improve predictive accuracy, interpretability decreases proportionally, creating an accuracy-interpretability trade-off [4]. The lack of transparency undermines trustworthiness when a human user inspects model output. In the context of residential load scheduling, this limitation is significant; automated decisions directly affect household electricity costs, and users require understandable justifications for recommended actions rather than just outputs alone. Despite growing recognition of this problem, interpretability remains underexplored in building energy management relative to predictive accuracy [4]. Post-hoc explainability techniques, SHAP in particular, have emerged as a means of addressing this gap by decomposing model predictions into individual feature contributions without sacrificing performance [5].

Large language models have recently emerged as a promising approach for generating natural language scheduling recommendations in Home Energy Management Systems (HEMS) [6]. The ability of LLMs to reason over input and produce plain-English justifications makes them appropriate for scheduling tasks, where explanations are important for the human user. However, existing LLM-based scheduling systems rely on the model's knowledge rather than grounding recommendations in data-driven evidence [6]. An LLM-based scheduling module that reasons without structured forecasting context may produce plausible but poorly grounded recommendations. This represents a significant gap in the current literature, one that motivates the integration of XAI-derived forecasting context as structured input to the LLM scheduling module.

A framework is proposed that bridges the gap by integrating SHAP-based explainability from a residential energy consumption forecasting model with a single LLM-based scheduling module. An XGBoost model [7] is trained on real residential energy consumption data from the PECAN Street Austin dataset [8] to generate 24-hour-ahead household grid consumption forecasts. SHAP TreeExplainer [5] is applied to the trained model to produce structured feature attribution summaries, identifying which input features drove the forecast above or below the baseline. These summaries are passed to Llama 3 [9] as decision context, alongside hourly forecast

values, solar generation data and real-time electricity pricing from the load zone. The module produces a 24-hour appliance schedule with plain-English justifications under two experimental conditions: Condition A, in which full SHAP context is provided, and Condition B, in which the same forecast and data are provided without SHAP feature attributions. This controlled comparison enables evaluation of whether SHAP context improves both performance and explanation quality.

The remainder of this paper is organised as follows. Section II reviews related work in explainable AI for energy forecasting and LLM-based home energy management. Section III describes the proposed methodology. Section IV presents the performance evaluation. Section V concludes the paper.

II. Related Work

A. Explainable AI for Energy Consumption Forecasting

Machine learning approaches, and gradient-boosted tree ensembles in particular, have demonstrated strong performance in residential energy consumption forecasting. Devanathan et al. propose an XGBoost-based framework with SHAP and LIME for multi-scale household forecasting, reporting that SHAP produces consistent global feature rankings with dominant predictors [2]. Bhandary et al. similarly apply ensemble methods with SHAP and LIME for household energy prediction, demonstrating that LIME supplements SHAP at the instance level for local explanation [3].

Van Zyl et al. conduct a comparative analysis of SHAP and Grad-CAM for feature selection in time-series energy forecasting, finding that SHAP-based feature engineering reduces model complexity while maintaining forecasting accuracy [10]. Darvishvand et al. provide a systematic review of XAI in urban building energy modelling, identifying SHAP as the dominant global and local interpretation method across the literature, while noting that evaluation of explanation quality remains largely qualitative [11]. A consistent limitation across these studies is that XAI techniques are applied post hoc to explain forecasting outputs but are not connected to downstream scheduling decisions.

B. LLM-based Home Energy Management Systems

Large language models have been explored as scheduling and decision support agents in HEMS. El Makroum et al. propose an agentic AI HEMS in which an LLM orchestrator coordinates specialist agents for individual appliances to retrieve day-ahead electricity prices and delegate scheduling tasks [6]. Zharova et al. present an explainable multi-agent recommendation system for energy-efficient decision support in smart homes to help improve transparency [1].

While these systems demonstrate scheduling performance, a noted limitation is the absence of structured explainability. In El Makroum et al., reasoning traces are developer-oriented and are not connected to the underlying forecasting evidence; the LLM reasons from pre-trained knowledge rather than specific patterns or prices [6]. No existing work connects SHAP feature attributions from a forecasting model to scheduling decisions of an LLM-based scheduling module in residential energy management. This gap motivates the framework in the present work.

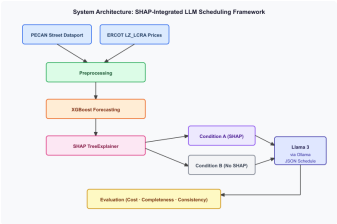


Fig. 1. System architecture of the proposed SHAP-integrated LLM scheduling framework.

C. XAI Integration and Evaluation in Energy Management

Beyond post-hoc explanation, recent work has researched the systematic integration of XAI in energy management. Clement et al. propose SHAP-based clustering of consumption patterns to detect behavioural shifts and support adaptive model retraining [12]. Beshaw et al. demonstrate SHAP-based transparency for users in static energy prediction tasks, finding that structured feature attribution improves user comprehension of model outputs [13]. Teixeira et al. present an explainable AI framework for automated energy management in buildings that integrates XAI outputs directly into operational decision support [14]. No study proposes a dual evaluation methodology that assesses scheduling performance and explanation quality with an LLM-based HEMS, a gap directly addressed by the evaluation design of the present work.

III. Methodology

A. Dataset and Preprocessing

Residential energy consumption data were obtained from the PECAN Street Dataport from households in Austin, Texas [8]. The 2018 dataset was selected, containing one-minute interval readings across 25 homes with 79 circuit-level columns per household. The year 2018 was selected as it provided complete annual coverage for home 4373 across all required columns and corresponded to the available ERCOT LZ_LCRA real-time pricing data, enabling temporally aligned consumption and pricing inputs. A single home, identified by dataport ID 4373, was selected for the prototype evaluation following a selection process. First, all 25 homes were assessed for data completeness across five columns: whole-home grid consumption, solar photovoltaic generation, washing machine, dishwasher, and electric vehicle charger.

In the next stage, EV charger activity was verified to confirm the presence of charging sessions. Home 2335, which showed 100% column completeness, was rejected after investigation, as its EV charger recorded only sensor noise (0.001–0.002 kW constant, zero minutes above 1 kW). Home 4373 was selected as it exhibited the highest real EV charging activity, with 64,787 one-minute readings above 1kW (consistent with a Level 2 home charger operating at approximately 3.3–3.9 kW).

Raw data were resampled from one-minute to hourly using mean aggregation, appropriate for power readings in kilowatts, where the mean represents average power over each hour. Negative appliance values, generally resulting from sensor noise, were clipped to zero, as negative power consumption is physically inadmissible for these circuits. Negative grid values were retained, as they represent legitimate solar export to the grid during periods of high photovoltaic generation. Real-time electricity pricing data from the ERCOT LZ_LCRA load zone for the year 2018 were merged with hourly consumption data, providing geographically aligned price signals in Austin. Prices were also converted from dollars per megawatt-hour to dollars per kilowatt-hour to maintain unit consistency with grid consumption measurements.

Temporal features were derived from local Austin time using the America/Chicago timezone, which correctly applies Central Standard Time (UTC-6) in winter and Central Daylight Time (UTC-5) in summer, as household consumption patterns are driven by local behaviour rather than UTC offsets. Raw timestamps were converted from UTC to local time before feature extraction, ensuring that the hour-of-day feature correctly reflects residential consumption patterns. Nine lag and rolling features were engineered to capture temporal autocorrelation: grid consumption lagged at 1, 2, 3, 24, and 168 hours, three-hour and 24-hour rolling means, and solar generation lagged at 1 and 24 hours. The first 168 rows were removed to eliminate missing values introduced by the weekly lag. The final preprocessed dataset comprised 8,592 hourly observations and 20 features for home 4373, covering the period January to December 2018. The evaluation is retrospective in nature, using historical consumption and pricing data to simulate scheduling decisions under perfect information conditions.

B. Energy Consumption Forecasting

An XGBoost model [7] was trained to forecast 24-hour-ahead household grid consumption for home 4373. XGBoost was selected over alternative models, such as LSTM networks, for two reasons motivated by the research objective. First, XGBoost achieves strong predictive performance on structured tabular data. Second, SHAP TreeExplainer is optimised for tree-based models, computing exact Shapley values efficiently using the tree structure. Since generating SHAP explanations is a primary objective of this work, the forecasting model was selected to maximise XAI compatibility.

Fifteen features were used as model inputs, comprising solar generation, electricity price, four temporal features (hour of day, day of week, month, weekend indicator), and nine lag and rolling features. Rolling average features were computed with a one-period shift to ensure only historical values contributed to each prediction, preventing potential target leakage. The three flexible appliance columns, washing machine, dishwasher, and EV charger, were deliberately excluded from the feature set to prevent data leakage. Including appliance consumption as a predictor of grid consumption would allow the model to learn a trivial mapping between appliance state and grid draw, producing inflated accuracy metrics that do not reflect genuine forecasting capability.

A chronological 80/20 train/test split was applied, with 6,873 observations used for training (January to October 2018) and 1,719 for testing (October to December 2018). Random splitting was not used as it would violate temporal ordering, allowing future lag values to appear in the training set and constituting data leakage. The exact split point was 2018-10-21 15:00 UTC, with training concluding at that timestamp and testing commencing immediately thereafter. The model was trained with 500 estimators, a learning rate of 0.05, a maximum tree depth of six, and a subsampling rate of 0.8. Model performance was evaluated using root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2) on the held-out test set.

C. SHAP Feature Attribution

SHAP TreeExplainer [5] was applied to the trained XGBoost model to generate feature attribution summaries for the 24-hour forecast window. SHAP assigns each input feature a Shapley value representing its marginal contribution to a given prediction relative to a baseline expected value. TreeExplainer was selected as it computes values by exploiting the tree structure of the model, making it theoretically sound and computationally efficient.

SHAP was applied to the test set of 1,719 observations, producing a matrix of 1,719 x 15 Shapley values. The expected value was 0.974kW, representing the baseline against which all feature contributions are measured. Global feature importance was computed as the mean absolute SHAP value across all test observations for each feature, providing a measure of each feature's average contribution to predictions across the full test period. Local SHAP analysis was conducted for each of the seven evaluation days spanning October 22nd to 28th 2018, the first complete week of the held-out test period, to ensure analysis reflects out-of-sample model behaviour.

For each day, the SHAP attribution summary was formatted as structured plain-text context comprising the 24-hour consumption forecast, top feature attributions with direction and magnitude, solar availability windows, lowest price hours, and observed appliance activity. In accordance with the scope of this work, SHAP explains the forecasting model only; the resulting feature attribution summary is passed to the LLM scheduling module as decision context, not as an explanation of the LLM scheduling module itself.

D. LLM Scheduling Module

A locally deployed LLM scheduling module was implemented using Llama 3 through Ollama to generate appliance scheduling recommendations within a household energy management context. The model was prompted with structured daily energy information, including predicted household load, electricity price, and available solar generation. Two prompt conditions were evaluated so that the influence of explainability information could be isolated while the underlying scheduling task remained constant. In Condition A, the prompt

included full SHAP-based feature attribution context, allowing the model to access ranked explanatory information about the forecast drivers alongside the energy data. In Condition B, the same core scheduling information was retained, but the SHAP-based features were removed. In this way, the effect of explanation context on model behaviour could be examined without altering the main task inputs.

The scheduling module was implemented using Llama 3 served locally via Ollama. Table I summarises the model configuration and hardware environment used for all inference runs.

TABLE I
LLM CONFIGURATION AND HARDWARE DETAILS

Parameter	Value
Model tag	`llama3:latest`
Parameter count	8.0B
Quantisation	Q4_0
Temperature	0.2
Context length	8,192 tokens
CPU	Intel Core i7-1355U (13th Gen)
RAM	15.7 GB
GPU	Intel Iris Xe (integrated)
Inference time	∼30 min (Cond. A); ∼25 min (Cond. B)

Temperature was set to 0.2 to promote deterministic, structured JSON output and reduce hallucinated formatting. All inference was performed on consumer laptop hardware with no discrete GPU, demonstrating the feasibility of locally-hosted LLM scheduling without cloud dependency.

The model was instructed to generate recommendations for three appliances in every case: the EV charger, washing machine, and dishwasher. For each appliance, the original operating time and a revised recommended time were required, together with a short justification of the change. To ensure consistency across both conditions, the output was constrained to a structured four-column JSON format containing Appliance, Original time, Recommended time, and Reason. This formatting choice was important because it allowed direct comparison between conditions, supported systematic analysis, and reduced ambiguity in the model outputs. The methodology therefore established a controlled framework in which scheduling behaviour and explanation quality could be evaluated side by side.

E. Evaluation Design

The evaluation framework assesses the proposed system under two experimental conditions across seven days spanning October 22nd to 28th 2018. Condition A provides the LLM-based scheduling module with the SHAP context consisting of the 24-hour forecast, feature attributions, solar availability, lowest price hours and appliance activity. Condition B provides the same forecast but omits the SHAP feature attribution. This controlled design ensures any observed difference in the scheduling performance or explanation is attributable to the presence of SHAP context.

Scheduling performance is evaluated using cost metrics derived from real-time ERCOT LZ_LCRA electricity prices. These wholesale market prices are used to simulate scheduling costs and do not represent actual retail tariffs paid by the household. For each appliance, a baseline cost is calculated as the electricity price at the original observed time multiplied by the assumed hourly energy consumption. The proposed cost is calculated using the LLM-based scheduling module recommended time. Cost savings are given as both absolute terms and as a percentage of the baseline cost. A positive saving indicates the LLM recommendation reduces electricity cost in relation to the non-scheduling baseline. Mean savings are reported per condition across all days and appliances.

Explanation quality is evaluated using two measurable proxies. First, completeness is assessed by checking whether the reason in each output references the top SHAP features identified in the global analysis. This check is performed programmatically by searching for relevant keywords in the reason text. Second, consistency is assessed by examining whether the LLM-based scheduling module produces similar recommendations across days with comparable solar and price values. Days are grouped by solar availability and price range, and outputs are compared for directional agreement.

The baseline against which both conditions are compared is defined as appliances operating at their original observed peak hours in the dataset for each day evaluated. This characterises the no-scheduling scenario: a household that does not adjust regardless of price or solar conditions. This baseline follows a similar approach to that of El Makroum et al. [6]

TABLE II
APPLIANCE ASSUMPTIONS USED IN COST CALCULATIONS

Appliance	Power (kW)	Duration (h)	Energy (kWh)[a]
EV charger	3.3–3.9	1	1
Washing machine	0.5	1	1
Dishwasher	1.2	1	1

[a]Energy consumption is set to 1 kWh for all appliances as a normalised assumption for cost comparison purposes.

IV. PERFORMANCE EVALUATION

A. Forecasting Model Performance

An XGBoost [7] model was trained on 6,873 hourly observations spanning January to October 2018 and evaluated on a held-out chronological test set of 1,719 observations covering October to December of the same year. A chronological rather than random split was applied to preserve temporal ordering and prevent data leakage through lag features. The model achieved a test set coefficient of determination (R^2) of 0.651, a root mean squared error (RMSE) of 0.996 kW, and a mean absolute error (MAE) of 0.643 kW, indicating moderate predictive performance relative to the mean household grid consumption of 0.880 kW.

Residual analysis suggested that prediction errors were approximately normally distributed and centred around zero, with no evidence of systematic heteroscedasticity across the range of predicted values, as illustrated in Fig. 2. A training R^2 of 0.974 compared to a test R^2 of 0.651 indicates a degree of overfitting consistent with gradient-boosted tree

models on tabular data; however, the test set performance confirms adequate generalisation for the purposes of SHAP feature attribution. These results indicate that the forecasting model produces adequate predictions suitable for generating meaningful SHAP feature attributions. Actual versus predicted values are shown in Fig. 3.

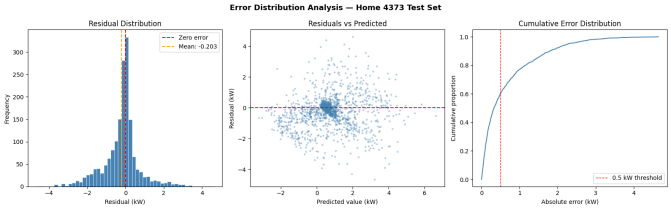


Fig. 2. Residual distribution and error analysis for the XGBoost forecasting model on the held-out test set.

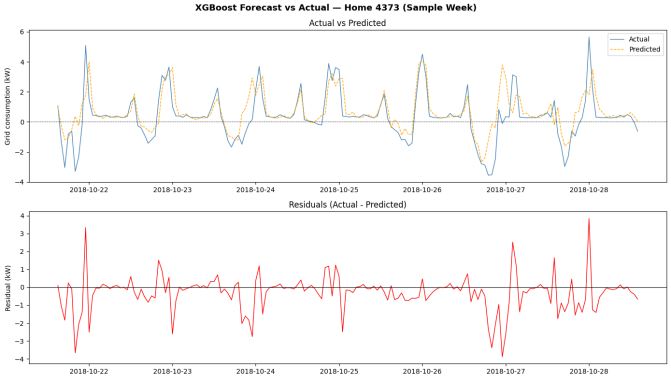


Fig. 3. Actual versus predicted grid consumption values on the held-out test set (Home 4373, October–December 2018).

Analysis of XGBoost built-in feature importance revealed that $grid_lag1$, the one-hour lagged grid consumption value, was the dominant predictor. The electricity price feature ($price_kwh$) ranked twelfth of fifteen features by XGBoost built-in feature importance, with near zero importance. This finding was subsequently confirmed by the SHAP global analysis and is consistent across both importance measures. It provides empirical evidence that the household does not adjust energy consumption in response to real-time electricity price signals, a consumption pattern that directly motivates the need for an active scheduling intervention.

B. SHAP Feature Attribution

SHAP TreeExplainer was applied to the trained XGBoost model across the full test set of 1,719 observations, producing a 1,719 x 15 matrix with the Shapley values. The expected value, representing the mean model prediction, was 0.974kW, against which all feature contributions are measured. Global feature importance, computed as the mean absolute SHAP value, indicated that $grid_lag1$ was the dominant predictor with a mean absolute SHAP value of 0.772 kW. Solar generation ranked second with a mean absolute SHAP value of 0.361 kW, with high solar values consistently producing negative SHAP contributions, confirming that the model correctly learned that solar generation reduces grid draw. The electricity price feature ranked thirteenth of fifteen features with a mean absolute SHAP value of 0.036 kW, validating the near-zero importance finding from the XGBoost built-in analysis and confirming this result is robust across both importance measures.

Local SHAP analysis was conducted across the seven evaluation days spanning October 22nd to 28th 2018. All 24 hourly SHAP attributions were verified to sum correctly to the model prediction for each day, confirming mathematical correctness of the implementation. The local analysis revealed that $grid_lag1$ SHAP values were strongly positive during high-consumption hours driven by EV charging activity, and reduced during periods when solar generation was highest and grid draw fell below the baseline expectation. Solar SHAP contributions were consistently negative during peak generation hours, correctly capturing the reduction in grid draw from photovoltaic output.

C. Scheduling Performance

Scheduling performance was evaluated by comparing appliance recommendations produced under both conditions against the no-scheduling baseline using real-time ERCOT LZ_LCRA electricity prices. Baseline costs were identical across both conditions ($0.718 total), confirming the controlled experimental design was maintained. Condition A achieved a mean cost saving of 71.0% relative to the baseline, improving on the baseline on all seven evaluation days. Excluding October 28th, where negative real-time prices produced anomalous savings figures, Condition A achieved a mean cost saving of 27.7% across the remaining six evaluation days. Condition B achieved a mean cost saving of 17.4%, improving on the baseline on five of seven days. These results indicate that SHAP-grounded context substantially improved scheduling performance across the evaluation period.

On five of seven days Condition A outperformed Condition B. The largest margin was observed on October 24th where Condition A achieved 57.2% mean savings compared to 25.6% for Condition B. On October 22nd both conditions performed comparably with a negligible difference in total savings. October 28th presented the most volatile price environment of the evaluation period, with prices ranging from -$1.01/MWh to $280.69/MWh. On this day, Condition A (with SHAP) demonstrated the value of grounded context by scheduling all three appliances into the negative-price evening window (hours 19:00-23:00 CDT), achieving a mean cost saving through the identification of negative real-time prices as the dominant scheduling signal. Condition B (without SHAP), lacking feature attribution context, scheduled the washing machine to 10:00 CDT, a $224.38/MWh hour immediately adjacent to the $280.69/MWh spike at 11:00 CDT, resulting in a cost increase of 143.5% on that appliance on that day. This contrast illustrates the benefit of SHAP context under high price volatility: the feature attribution summary drew attention to price as a near-zero-importance driver of passive

consumption, motivating more aggressive active scheduling toward low-price and negative-price hours.

TABLE III
CONDITION A VS. CONDITION B SUMMARY

Condition	Mean Savings (%)	Completeness	Consistency (%)
Cond. A (with SHAP)	71.0	2.14/4	76.2
Cond. B (no SHAP)	17.4	1.90/4	61.9

D. Explanation Quality

Explanation quality was evaluated by examining how completely and consistently the model justified its scheduling recommendations under both prompt conditions. Completeness was assessed by checking whether reasons referenced the top SHAP features identified in the global analysis, specifically $grid_lag1$, solar generation, time-of-day patterns, and electricity price. Condition A achieved a mean completeness score of 2.14 out of 4 compared to 1.90 out of 4 for Condition B, indicating that SHAP context produced more complete and better-grounded explanations. The most notable difference was in references to consumption lag patterns, $grid_lag1$ was mentioned in 8 Condition A recommendations compared to 3 in Condition B, a finding directly attributable to the explicit SHAP feature attribution provided in the Condition A prompt. References to solar generation and price were comparable across both conditions, as both received the same forecast and solar data.

Consistency was assessed by examining whether the LLM-based scheduling module produced directionally similar recommendations across the seven evaluation days. Condition A achieved a consistency score of 76.2% compared to 61.9% for Condition B, indicating that SHAP context produced more predictable and stable recommendations. In Condition A the washing machine was recommended to shift to a later hour on all seven days, while Condition B showed greater directional variability across appliances. Overall, SHAP context produced explanations that more frequently referenced the supplied forecast features, achieving a higher keyword completeness score and greater directional consistency in recommendations than the no-SHAP baseline condition. Semantic correctness and user comprehensibility were not formally evaluated.

V. CONCLUSION

This paper presented a framework integrating SHAP-based explainability for an energy consumption forecasting model with an LLM-based scheduling module for household load scheduling. An XGBoost model trained on energy data from the PECAN Street Austin dataset achieved a test R^2 of 0.651 and an RMSE of 0.996 kW, confirming adequate predictive performance of the forecasting foundation. SHAP TreeExplainer was applied to generate structured feature attribution summaries identifying the dominant drivers of household consumption forecasts, which were then passed to Llama 3 as decision context under two experimental conditions. The evaluation demonstrated that SHAP context produced explanations that more frequently referenced the supplied forecast features, with Condition A achieving a higher keyword completeness score. A key finding from the forecasting analysis was that electricity price ranked thirteenth of fifteen features in SHAP global importance, providing evidence that passive households do not respond to real-time price signals and that active scheduling is required to realise demand-side flexibility potential. Scheduling performance demonstrated improvement under Condition A, which achieved a mean cost saving of 71.0% and improved on the no-scheduling baseline on all seven evaluation days. October 28th, where prices ranged from -$1.01/MWh to $280.69/MWh, demonstrated the particular value of SHAP grounding under high price volatility: Condition A correctly identified the negative-price evening window while Condition B, lacking feature attribution context, scheduled the washing machine into a high-price window, incurring a cost increase of 143.5% on that appliance.

Several limitations of the prototype evaluation should be acknowledged. First, the study was conducted on a single household over a seven-day evaluation period, which is insufficient for statistical generalisation; results should be interpreted as indicative findings of a prototype system rather than definitive conclusions. Second, the cost calculation assumed each appliance ran for one hour at its observed peak power, simplifying the real energy consumption profiles of appliances that operate over variable durations. Third, the explanation completeness metric relied on keyword matching against a predefined set of SHAP-relevant terms, which cannot capture semantic nuance or assess the logical correctness of natural language explanations. Fourth, the current SHAP context format pre-computes solar availability windows and lowest price hours as aggregate summaries, but does not explicitly flag extreme intra-day price anomalies. This limitation contributed to Condition B recommending the washing machine to a $224.38/MWh hour immediately adjacent to the $280.69/MWh spike on 28th October 2018, prioritising solar availability over price avoidance. Fifth, the LLM was deployed at a low temperature (0.2), which is near-deterministic but limits exploration of output diversity and may not reflect real-world deployment conditions where variability in recommendations could be beneficial.

Several potential research directions are identified. The framework should be extended to multiple homes and longer evaluation periods to assess cross-household generalisability and robustness. Prompt engineering improvements may substantially improve scheduling performance on volatile price days, directly addressing the limitation identified on 28th October 2018. The explanation quality evaluation could be strengthened through the use of semantic similarity metrics or human evaluation studies to assess whether SHAP-grounded explanations are genuinely more useful to end users than generic off-peak reasoning. Finally, the framework could be extended to incorporate real-time SHAP updates as new consumption data becomes available throughout the day, enabling adaptive rather than static 24-hour scheduling.

TABLE IV
FULL SEVEN-DAY SCHEDULING RESULTS — OCTOBER 22–28 2018

Date	Appliance	Orig.	Cond. A	Cond. B	Baseline ($/kWh)	A Saving (%)	B Saving (%)
2018-10-22	EV charger	15:00	14:00	14:00	0.0206	-4.78	0.00
	Washing machine	08:00	18:00	06:00	0.0239	20.99	-13.79
	Dishwasher	23:00	20:00	22:00	0.0208	8.72	6.04
2018-10-23	EV charger	07:00	09:00	19:00	0.0248	19.46	-25.91
	Washing machine	12:00	16:00	21:00	0.0229	6.81	-18.09
	Dishwasher	22:00	20:00	18:00	0.0191	5.24	6.00
2018-10-24	EV charger	15:00	21:00	14:00	0.0463	-35.45	0.00
	Washing machine	11:00	22:00	20:00	0.0390	54.86	55.00
	Dishwasher	15:00	20:00	21:00	0.0463	62.02	60.89
2018-10-25	EV charger	18:00	16:00	16:00	0.0754	64.39	64.39
	Washing machine	13:00	22:00	20:00	0.0497	7.43	39.13
	Dishwasher	01:00	21:00	21:00	0.0338	32.20	32.20
2018-10-26	EV charger	18:00	16:00	16:00	0.0398	45.00	45.00
	Washing machine	13:00	14:00	04:00	0.0398	22.60	12.95
	Dishwasher	02:00	20:00	20:00	0.0284	26.88	26.88
2018-10-27	EV charger	13:00	17:00	16:00	0.0318	52.61	47.14
	Washing machine	11:00	20:00	04:00	0.0408	57.70	56.81
	Dishwasher	08:00	23:00	20:00	0.0393	51.05	55.70
2018-10-28	EV charger	18:00	19:00	19:00	0.0003	433.67	433.67
	Washing machine	09:00	22:00	10:00	0.0921	100.23	-143.5
	Dishwasher	18:00	20:00	18:00	0.0003	367.58	0.00

GENERATIVE AI DISCLOSURE

Generative AI tools, specifically Claude (Anthropic), were used during the preparation of this work to assist with code development, writing support, and review of draft sections. AI assistance was used across the following components: Python code for data preprocessing, model training, SHAP analysis, and evaluation notebooks; review and providing feedback on drafts throughout the writing process. All experimental design decisions, data processing choices, model selection, evaluation methodology, and research conclusions were made by the authors. All code was executed, verified, and validated by the authors on real data. Any written content was reviewed, edited, and approved by the authors. The Llama 3 language model was used as the scheduling module within the experimental framework described in this paper and is cited accordingly as a research tool.

REFERENCES

[1] A. Zharova, A. Boer, J. Knoblauch, Kai Ingo Schewina, and J. Vihs, "An explainable multi-agent recommendation system for energy-efficient decision support in smart homes," Environmental data science, vol. 3, Jan. 2024, doi: 10.1017/eds.2024.8.

[2] B. Devanathan, K. Jnana Varshitha, L. P. Kumar, S. A. Lakshmanan, and N. K. Prakash, "Explainable AI Framework Using XGBoost with SHAP and LIME for Multi-Scale Household Energy Forecasting.," IEEE Access, pp. 1–1, Jan. 2025, doi: 10.1109/access.2025.3602673.

[3] Aakash Bhandary, Vruti Dobariya, Gokul Yenduri, R. H. Jhaveri, Saikat Gochhait, and F. Benedetto, "Enhancing Household Energy Consumption Predictions Through Explainable AI Frameworks," IEEE Access, vol. 12, pp. 36764–36777, Jan. 2024, doi: 10.1109/access.2024.3373552.

[4] L. Gugliermetti, F. Cumo, and S. Agostinelli, "A Future Direction of Machine Learning for Building Energy Management: Interpretable Models," Energies, vol. 17, no. 3, p. 700, Feb. 2024, doi: 10.3390/en17030700.

[5] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," Neural Information Processing Systems, 2017. https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html (accessed July 18, 2026).

[6] R. El Makroum, S. Zwickl-Bernhard, and L. Kranzl, "Agentic AI Home Energy Management System: A Large Language Model Framework for Residential Load Scheduling," arXiv.org, 2025. https://arxiv.org/abs/2510.26603 (accessed July 18, 2026).

[7] T. Chen and C. Guestrin, "XGBoost: a Scalable Tree Boosting System," Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16, vol. 1, no. 1, pp. 785–794, Aug. 2016, doi: 10.1145/2939672.2939785.

[8] "Dataport," Pecan Street Inc. https://www.pecanstreet.org/dataport/ (accessed July 18, 2026).

[9] A. Dubey et al., "The Llama 3 Herd of Models," arXiv.org, 2024. https://arxiv.org/abs/2407.21783 (accessed July 18, 2026).

[10] Corne van Zyl, X. Ye, and R. Naidoo, "Harnessing eXplainable artificial intelligence for feature selection in time series energy forecasting: A comparative analysis of Grad-CAM and SHAP," Applied energy, vol. 353, pp. 122079–122079, Jan. 2024, doi: 10.1016/j.apenergy.2023.122079.

[11] L. Darvishvand, B. Kamkari, M. J. Huang, and N. J. Hewitt, "A systematic review of explainable artificial intelligence in urban building energy modeling: methods, applications, and future directions," Sustainable Cities and Society, vol. 128, p. 106492, June 2025, doi: 10.1016/j.scs.2025.106492.

[12] T. Clement, H. T. T. Nguyen, N. Kemmerzell, M. Abdelaal, and D. Stjelja, "Beyond explaining: XAI-based Adaptive Learning with SHAP Clustering for Energy Consumption Prediction," arXiv.org, 2024. https://arxiv.org/abs/2402.04982 (accessed July 18, 2026).

[13] F. G. Beshaw, Thamir Hassan Atyia, Mohd, M. K. Ishak, and A. S. Din, "Utilizing Machine Learning and SHAP Values for Improved and Transparent Energy Usage Predictions," Computers, materials & continua/Computers, materials & continua (Print), vol. 0, no. 0, pp. 1–10, Jan. 2025, doi: 10.32604/cmc.2025.061400.

[14] B. Teixeira, L. Carvalhais, T. Pinto, and Z. Vale, "Explainable AI framework for reliable and transparent automated energy management in buildings," Energy and Buildings, vol. 347, p. 116246, Aug. 2025, doi: 10.1016/j.enbuild.2025.116246.